Rapid response to radiation of choroidal melanoma

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Numerous investigators have documented regression of uveal melanomas after various methods of radiotherapy. The definition of "successful result" has varied from complete regression to a flat scar to any degree of measurable regression. Stallard[1] stated in 1966 that failure of treated tumours to regress completely should be regarded as an "uncertain result" and suggested that such tumours required close follow-up for any signs of activity. Other investigators have studied the relation between rate of tumour regression and patient survival and observed a higher rate of death from metastatic melanoma among patients whose tumours regressed rapidly following brachytherapy.[2,3] We report the case of a patient with choroidal melanoma that had regressed from 6.2 mm in thickness to almost a flat scar 10 weeks after iodine 125 brachytherapy, with subsequent tumour recurrence.

Case report

A 58-year-old white woman was referred to the Ocular Oncology Service for evaluation of a possible choroidal tumour in her right eye. She had noted blurred vision for the previous 4 months. Examination showed a corrected visual acuity of 20/200 in the right eye and 20/15 in the left. The anterior segments were unremarkable. Funduscopy of the right eye with the pupil dilated revealed an amelanotic mass arising from the superotemporal fundus measuring 15 mm by 13 mm at the base (Fig. 1, A). The lesion had involved the macula and was associated with inferior serous retinal detachment. Fluorescein angiography showed tumour circulation in the early choroidal phase. B-scan ultrasonography demonstrated a dome-shaped solid tumour measuring 6.2 mm in height (Fig. 1, B), with low internal reflectivity on the A-scan. Metastatic investigation yielded negative results. Choroidal melanoma was diagnosed, and ^{125}I brachytherapy was performed 1 month after presentation.

Ten weeks later the woman's visual acuity had improved to 20/60, and the tumour had regressed to 1.2 mm in height, leaving a slightly elevated and mottled chorioretinal scar (Fig. 1, C). There was total resolution of the subretinal fluid. Two months later the visual acuity was 20/30, and the chorioretinal scar measured 0.7 mm in thickness (Fig. 1, D).

A follow-up examination 1 year later showed a new, mildly pigmented, elevated lesion along the nasal margin of the previously treated tumour (Fig. 1, E). The lesion had indistinct margins and measured 1.4 mm in thickness, consistent with tumour recurrence. Metastatic investigation remained negative. Transpupillary thermotherapy was applied to the recurrent tumour, which continued to grow, and the eye was eventually enucleated 5½ months after the follow-up examination. Pathological examination revealed mixed-cell melanoma with invasion into an emissary canal; however, there was no evidence of extrascleral extension.

Comments

The key to any treatment regimen for ocular melanoma must be local tumour control and patient survival. Studies of tumour radiation biology suggest that the radioresponsiveness of a gross tumour reflects the radiosensitivity of its constituent cells.[4] Iodine 125 is a low-energy source that allows adequate irradiation of tumours up to 10 mm in thickness. The success of radiotherapy of a choroidal melanoma depends on even irradiation throughout the tumour as well as an adequate integral dose. The seed array in the gold plaque is important for an even distribution of the radiation throughout the tumour. In our patient a radiation dose of 100 cGy was delivered to the tumour apex over a 6-day exposure interval at a rate of 100 cGy/h. The dose rate is critical, since melanoma cells have a

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